H524 Introduction to Biostatistics
Week 3 Assignment - ANSWER KEY
Fall 2025

Total Points: 100 (+ 5 bonus)
Instructor Use Only

Multiple Choice Answer Key

Part 1: Binomial Distribution (23 points)

Q#	Answer	Pts	Topic
1	A	5	Binomial conditions
2	A	3	$P(X=12) = 0.2501$
3	B	3	$P(X \geq 13) = 0.3980$
4	A	2	R code for binomial
5	A	2	Expected value = 12
6	A	2	SD = 1.549
7	A	3	Vaccine $P(X<20) = 0.0334$
8	A	3	Vaccine $P(22 \leq X \leq 24) = 0.6918$

Part 2: Normal Distribution (22 points)

Q#	Answer	Pts	Topic
9	A	2	Z-score = 1.5
10	A	3	$P(X>185) = 0.0668$
11	A	3	$P(165<X<180) = 0.5328$
12	A	3	90th percentile = 182.8 cm
14	A	2	25th percentile = 163.3 cm
16	A	2	Empirical rule: 120 to 280 mg/dL
17	A	3	$P(X>280) = 0.0228$
18	A	2	$P(BP>140) = 0.2024$
19	A	2	Expected hypertensive = 202

Part 3: Sampling Distributions and CLT (23 points)

Part 4: CI Interpretation (5 points)

Q#	Answer	Pts	Topic
20	A	2	$E(\bar{X}) = 100$
21	A	2	SE = 4
22	A	3	SE with n=100 is 2
23	A	2	CLT: $N(100, 3.125^2)$
24	A	4	P($\bar{X}$>105) = 0.0548
25	A	2	P(97<$\bar{X}$<103) = 0.6629
26	A	2	SE = 5 (with σ=30, n=36)
27	A	2	SE decreases with larger n
28	A	1	SD vs SE concept
29	A	3	P($\bar{X}$<3350) = 0.1587
30	A	2	R code for CLT

Q#	Answer	Pts	Topic
31	A	2	Correct CI interpretation
32	A	2	Meaning of 95% confidence
33	A	1	99% CI is wider

Multiple Choice - Detailed Solutions

Part 1: Binomial Distribution

Question 2: P(X = 12) for n=15, p=0.80
Solution:

$$P(X = 12) = \binom{15}{12}(0.80)^{12}(0.20)^3 = \mathbf{0.2501}$$

R code: `dbinom(12, 15, 0.80)`

Question 3: P(X $\geq$ 13) for n=15, p=0.80
Solution:

$$P(X \geq 13) = P(X = 13) + P(X = 14) + P(X = 15) = \mathbf{0.3980}$$

R code: `sum(dbinom(13:15, 15, 0.80))` or `pbinom(12, 15, 0.80, lower.tail=FALSE)`

Question 5: Expected value for n=15, p=0.80
Solution:

$$E(X) = np = 15 \times 0.80 = \mathbf{12}$$

Question 6: Standard deviation for n=15, p=0.80
Solution:

$$SD(X) = \sqrt{np(1-p)} = \sqrt{15 \times 0.80 \times 0.20} = \sqrt{2.4} = \mathbf{1.549}$$

Question 7: Vaccine efficacy - P(X < 20) for n=25, p=0.90
Solution:

$$P(X < 20) = P(X \leq 19) = \mathbf{0.0334}$$

R code: `pbinom(19, 25, 0.90)`

Question 8: Vaccine efficacy - P(22 $\leq$ X $\leq$ 24) for n=25, p=0.90
Solution:

$$P(22 \leq X \leq 24) = P(X = 22) + P(X = 23) + P(X = 24) = \mathbf{0.6918}$$

R code: `sum(dbinom(22:24, 25, 0.90))` or `pbinom(24, 25, 0.90) - pbinom(21, 25, 0.90)`

Part 2: Normal Distribution

Question 9: Z-score for X=185, μ=170, σ=10
Solution:

$$Z = \frac{X - \mu}{\sigma} = \frac{185 - 170}{10} = \frac{15}{10} = \mathbf{1.5}$$

Question 10: P(X > 185) for μ=170, σ=10
Solution:

$$Z = 1.5 \text{ (from Q9)}$$

$$P(X > 185) = P(Z > 1.5) = \mathbf{0.0668}$$

R code: `pnorm(185, 170, 10, lower.tail=FALSE)` or `1 - pnorm(1.5)`

Question 11: P(165 < X < 180) for μ=170, σ=10
Solution:

$$Z_1 = \frac{165 - 170}{10} = -0.5, \quad Z_2 = \frac{180 - 170}{10} = 1.0$$

$$P(165 < X < 180) = P(-0.5 < Z < 1.0) = \mathbf{0.5328}$$

R code: `pnorm(180, 170, 10) - pnorm(165, 170, 10)`

Question 12: 90th percentile for μ=170, σ=10
Solution:

$$Z_{0.90} = 1.282 \text{ (from z-table or R)}$$

$$X = \mu + Z\sigma = 170 + 1.282(10) = \mathbf{182.8 \ cm}$$

R code: `qnorm(0.90, 170, 10)`

Question 14: 25th percentile for μ=170, σ=10
Solution:

$$Z_{0.25} = -0.674 \text{ (from z-table or R)}$$

$$X = \mu + Z\sigma = 170 + (-0.674)(10) = \mathbf{163.3 \ cm}$$

R code: `qnorm(0.25, 170, 10)`

Question 16: Empirical rule - 95% range for μ=200, σ=40
Solution:

$$95\% \text{ of data: } \mu \pm 2\sigma = 200 \pm 2(40) = \mathbf{120 \ to \ 280 \ mg/dL}$$

Question 17: P(X > 280) for μ=200, σ=40
Solution:

$$Z = \frac{280 - 200}{40} = 2.0$$

$$P(X > 280) = P(Z > 2.0) = \mathbf{0.0228}$$

R code: `pnorm(280, 200, 40, lower.tail=FALSE)`

Question 18: P(BP > 140) for μ=125, σ=18
Solution:

$$Z = \frac{140 - 125}{18} = 0.833$$

$$P(X > 140) = P(Z > 0.833) = \mathbf{0.2024}$$

R code: `pnorm(140, 125, 18, lower.tail=FALSE)`

Question 19: Expected number hypertensive in 1000 people
Solution:

$$\text{Expected} = n \times p = 1000 \times 0.2024 = \mathbf{202}$$

Part 3: Sampling Distributions and CLT

Question 20: Mean of sampling distribution, μ=100, σ=20, n=25
Solution:

$$E(\bar{X}) = \mu = \mathbf{100}$$

Question 21: Standard error, μ=100, σ=20, n=25
Solution:

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{20}{\sqrt{25}} = \frac{20}{5} = \mathbf{4}$$

Question 22: Standard error with n=100
Solution:

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{20}{\sqrt{100}} = \frac{20}{10} = \mathbf{2}$$

Question 23: CLT distribution, μ=100, σ=25, n=64
Solution:

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{25}{\sqrt{64}} = \frac{25}{8} = 3.125$$

$$\bar{X} \sim N(100, 3.125^2)$$

Question 24: P($\bar{X}$ > 105), μ=100, SE=3.125
Solution:

$$Z = \frac{105 - 100}{3.125} = 1.6$$

$$P(\bar{X} > 105) = P(Z > 1.6) = 0.0548$$

R code: `pnorm(105, 100, 3.125, lower.tail=FALSE)`

Question 25: P($97 < \bar{X} < 103$), μ=100, SE=3.125
Solution:

$$Z_1 = \frac{97-100}{3.125} = -0.96, \quad Z_2 = \frac{103-100}{3.125} = 0.96$$

$$P(97 < \bar{X} < 103) = P(-0.96 < Z < 0.96) = 0.6629$$

R code: `pnorm(103, 100, 3.125) - pnorm(97, 100, 3.125)`

Question 26: Standard error, σ=30, n=36
Solution:

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{30}{\sqrt{36}} = \frac{30}{6} = 5$$

Question 29: P($\bar{X} < 3350$), μ=3400, σ=500, n=100
Solution:

$$SE = \frac{500}{\sqrt{100}} = 50$$

$$Z = \frac{3350-3400}{50} = -1.0$$

$$P(\bar{X} < 3350) = P(Z < -1.0) = 0.1587$$

R code: `pnorm(3350, 3400, 50)`

Fill-In Questions - Detailed Solutions

Question 13: Manual CI Calculation (10 points)

Data: $\bar{x} = 6.2$, $s = 2.4$, $n = 20$

Part a: (2 points) Should you use a z or t distribution? Why?

Answer: Use **t-distribution** because:

- Population SD (σ) is unknown
- Small sample size ($n = 20 < 30$)

Part b: (2 points) Find the appropriate critical value for a 95% confidence interval.

Answer:

- $df = n - 1 = 19$
- $t_{0.975,19} = \mathbf{2.093}$
- R code: `qt(0.975, 19)`

Part c: (2 points) Calculate the standard error.

Answer:

$$SE = \frac{s}{\sqrt{n}} = \frac{2.4}{\sqrt{20}} = \frac{2.4}{4.472} = \mathbf{0.536}$$

Part d: (2 points) Calculate the margin of error.

Answer:

$$ME = t_{critical} \times SE = 2.093 \times 0.536 = \mathbf{1.122}$$

Part e: (2 points) Construct the 95% confidence interval.

Answer:

$$CI = \bar{x} \pm ME = 6.2 \pm 1.122 = \mathbf{(5.08,\ 7.32)} \quad \textbf{points}$$

Question 15: CI with R (5 points)

Data: Cholesterol reduction for 30 patients

Part a: (2 points) Write R code to calculate a 95% confidence interval using `t.test()`.

Answer:

```
cholesterol_reduction <- c(28, 32, 25, 30, 27, 35, 22, 29, 31, 26,
                           24, 33, 28, 30, 27, 29, 31, 25, 28, 32,
                           26, 30, 29, 27, 34, 28, 31, 25, 29, 30)

result <- t.test(cholesterol_reduction, conf.level = 0.95)
result$conf.int
```

Part b: (1 point) Report the confidence interval.

Answer: 95% CI: **(27.57, 29.83) mg/dL**

Part c: (2 points) Based on this interval, is there strong evidence that the mean reduction exceeds 25 mg/dL? Explain.

Answer: **Yes**, there is strong evidence. The entire 95% confidence interval is above 25 mg/dL, so we can be 95% confident that the true mean exceeds 25 mg/dL.

Question 34: Comprehensive Problem (5 points)

Data: 80 patients, 72 successful, $\bar{x} = 35$ mg/dL, $s = 8$ mg/dL

Part a: (2 points) Calculate a 95% confidence interval for the true success rate (proportion).

Answer:

$$\hat{p} = \frac{72}{80} = 0.90$$
$$SE_p = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.90(0.10)}{80}} = 0.0335$$
$$CI = 0.90 \pm 1.96(0.0335) = 0.90 \pm 0.066 = \mathbf{(0.834,\ 0.966)}$$
$$\text{or } \mathbf{(83.4\%,\ 96.6\%)}$$

Part b: (2 points) Calculate a 95% confidence interval for the true mean cholesterol reduction.

Answer:

$$SE = \frac{s}{\sqrt{n}} = \frac{8}{\sqrt{80}} = 0.894$$
$$CI = 35 \pm 1.96(0.894) = 35 \pm 1.752 = \mathbf{(33.25,\ 36.75)} \quad \mathbf{mg/dL}$$

Part c: (1 point) Write complete R code to perform both calculations and create a visualization.

Answer:

```r
# Part a: CI for proportion
n <- 80
successes <- 72
p_hat <- successes / n
se_p <- sqrt(p_hat * (1 - p_hat) / n)
ci_prop <- p_hat + c(-1, 1) * 1.96 * se_p
cat("95% CI for success rate:", round(ci_prop, 3), "\n")

# Part b: CI for mean
chol_mean <- 35
chol_sd <- 8
se_mean <- chol_sd / sqrt(n)
ci_mean <- chol_mean + c(-1, 1) * 1.96 * se_mean
cat("95% CI for mean:", round(ci_mean, 2), "mg/dL\n")

# Visualization
set.seed(123)
chol_data <- rnorm(80, mean = 35, sd = 8)
hist(chol_data, main = "Cholesterol Reduction Distribution",
     xlab = "Reduction (mg/dL)", col = "lightblue", breaks = 15)
abline(v = ci_mean, col = "red", lwd = 2, lty = 2)
legend("topright", legend = "95% CI", col = "red", lty = 2, lwd = 2)
```

Question 35: AI Verification and Learning (5 bonus points)

Grading Notes:

Part a: (1 point) Student should provide actual AI prompt and response
Part b: (2 points) Student should verify AI answer step-by-step, identifying any errors

- Step-by-step verification: 1 point
- Correctly identifying accuracy/errors: 1 point

Part c: (2 points) Student should evaluate quality of AI's CLT explanation

- Providing AI's explanation: 0.5 points
- Thoughtful evaluation of quality: 1 point
- Discussion of strengths/weaknesses: 0.5 points

Look for:

- Clear documentation of AI interaction
- Critical thinking about AI responses
- Understanding of statistical concepts
- Ability to verify and evaluate AI output

Quick Reference: Correct Answers Summary

All MC Questions (31 total): Q1-Q12, Q14, Q16-Q33 - Answer is always A for this version

Fill-In Questions (4 total):

- Q13 (10 pts): Manual CI = (5.08, 7.32)
- Q15 (5 pts): R-based CI = (27.57, 29.83); Yes, exceeds 25
- Q34 (5 pts): Proportion CI = (83.4%, 96.6%); Mean CI = (33.25, 36.75)
- Q35 (5 pts bonus): Graded based on quality of AI interaction and analysis

Grading Summary

Section	**Points**	**Score**
Part 1: Binomial Distribution (Q1-Q8)	23	
Part 2: Normal Distribution (Q9-Q12, Q14, Q16-Q19)	22	
Part 3: Sampling/CLT (Q20-Q30)	23	
Part 4: Manual CI (Q13)	10	
Part 4: CI Interpretation (Q31-Q33)	5	
Part 4: CI with R (Q15)	5	
Part 5: Comprehensive (Q34)	5	
Part 5: Formulas/Tables	7	
Subtotal	**100**	
Bonus: AI Learning (Q35)	+5	
Total Possible	**105**	

Note: This answer key reflects the updated assignment with 31 multiple choice questions and 4 fill-in questions (including 1 bonus).